

Random Exercises

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1 Overview

The following document is a compilation of various calculations, computations, and exercises I completed as part of orals prep. The work here is not meant to be an extensive account of my work, by any means.

2 Evans

Problem 2.18. (Stokes' rule) Assume u solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = 0, u_t = h, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Show that $v := u_t$ solves

$$\begin{cases} v_{tt} - \Delta v = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ v = h, v_t = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

This is *Stokes' rule*.

Assume u solves the given IVP, and let $v := u_t$. One immediately notes that on $\mathbb{R}^n \times \{t = 0\}$, since $u_t = h$, $v = h$. Second, $v_t = u_{tt}$. On $\mathbb{R}^n \times \{t = 0\}$, $u \equiv 0$ so that $\Delta u = 0$. But since u satisfies the wave equation $u_{tt} = \Delta u \equiv 0$. Thus $v_t \equiv 0$ on $\mathbb{R}^n \times \{t = 0\}$. Next we note that ∂_t and ∂_{x^j} commute for any $1 \leq j \leq n$. Thus since u satisfies $u_{tt} - \Delta u = 0$, differentiating both sides by t , one obtains

$$u_{ttt} - \Delta u_t = 0 \iff v_{tt} - \Delta v = 0$$

in $\mathbb{R}^n \times (0, \infty)$.

Problem 2.19.

(a) Show the general solution of the PDE $u_{xy} = 0$ is

$$u(x, y) = F(x) + G(y)$$

for arbitrary functions F, G .

(b) Using the change of variables $\xi = x + t$, $\eta = x - t$, show $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$.

(c) Use (a) and (b) to rederive d'Alembert's formula.

(d) Under what conditions on the initial data g, h is the solution u a right-moving wave? A left-moving wave?

(a) Consider the PDE $u_{xy} = 0$. Let $v(x, y) = u_x$. Then v satisfies the differential equation $v_y = 0$. Thus since v is independent of y , one has $u_x(x, y) = v(x, y) = f(x)$ for some arbitrary function f . But then integrating both sides with respect to x , and supposing that $F(x)$ is the antiderivative of $f(x)$,

$$u(x, y) = F(x) + G(y),$$

where G is some arbitrary function of y .

(b) Define $\xi = x + t$ and $\eta = x - t$ so that $x = (\xi + \eta)/2$ and $t = (\xi - \eta)/2$. Then,

$$\begin{aligned} u_\xi &= u_x \frac{\partial x}{\partial \xi} + u_t \frac{\partial t}{\partial \xi} \\ &= \frac{1}{2} u_x + \frac{1}{2} u_t. \\ u_{\xi\eta} &= \frac{1}{2} \left[u_{xx} \frac{\partial x}{\partial \eta} + u_{xt} \frac{\partial t}{\partial \eta} \right] + \frac{1}{2} \left[u_{tx} \frac{\partial x}{\partial \eta} + u_{tt} \frac{\partial t}{\partial \eta} \right] \\ &= \frac{1}{2} [u_{xx} - u_{xt}] + \frac{1}{2} [u_{tx} - u_{tt}] = \frac{1}{2} [u_{tt} - u_{xx}], \end{aligned}$$

where we used the fact that $u_{xt} = u_{tx}$. Thus, the proof concludes.

(c) Consider the initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Define the change of variables $\xi = x + t$ and $\eta = x - t$ as above. By (b), $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$. By (a), it follows that u must be of the form,

$$u(\xi, \eta) = F(\xi) + G(\eta).$$

Thus, $u(x, t) = F(x + t) + G(x - t)$. Now we must match the initial conditions. At $t = 0$,

$$u(x, 0) = F(x) + G(x) = g(x).$$

$$u_t(x, 0) = F'(x) - G'(x) = h(x).$$

Differentiating the first equation, one first obtains the system

$$\begin{cases} F'(x) + G'(x) = g'(x) \\ F'(x) - G'(x) = h(x). \end{cases}$$

Thus, we get

$$F'(x) = \frac{1}{2}(g'(x) + h(x)) \implies F(x + t) = \frac{1}{2}[g(x + t) - g(0)] + \frac{1}{2} \int_0^{x+t} h(y) dy + F(0).$$

Likewise, we get

$$G(x - t) = \frac{1}{2}[g(x - t) - g(0)] - \int_0^{x-t} h(y) dy + G(0).$$

Adding the two equations, and using the fact that $F(0) + G(0) = g(0)$, we conclude that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy,$$

which is precisely d'Alembert's formula.

- (d) To get a left-moving wave, we want the functions of $(x - t)$ to cancel out. Suppose $h(x) = g'(x)$. Then the fundamental theorem of calculus implies that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} g'(y) dy = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2}[g(x + t) - g(x - t)] = g(x + t).$$

Likewise, to get a right-moving wave, we want the functions of $(x + t)$ to cancel out, so we can take $h(x) = -g'(x)$.

Problem 2.21. (Equipartition of Energy). Let $u \in C^2(\mathbb{R} \times [0, \infty))$ solve the initial-value problem for the wave equation in one-dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Suppose g, h have compact support. The *kinetic energy* is $k(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_t^2(x, t) dx$ and the *potential energy* is $p(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_x^2(x, t) dx$. Prove

- (i) $k(t) + p(t)$ is constant in t ,
- (ii) $k(t) = p(t)$ for all large enough times t .

- (i) Let us define the kinetic and potential energy, $k(t)$ and $p(t)$ as given above. We wish to show that the total energy

$$e(t) = k(t) + p(t) = \frac{1}{2} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx$$

is conserved, where u is given by d'Alembert's formula,

$$u(x, t) = \frac{1}{2}[g(x + t) + g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy.$$

so that

$$\begin{aligned} u_t(x, t) &= \frac{1}{2}[g'(x+t) - g'(x-t) + h(x+t) + h(x-t)], \\ u_x(x, t) &= \frac{1}{2}[g'(x+t) + g'(x-t) + h(x+t) - h(x-t)]. \end{aligned}$$

Let $T > 0$, and let $K \subset \mathbb{R}$ be a compact set large enough so that $\text{supp } g, \text{supp } h$ are both contained in K . Then it follows that u_t^2 and u_x^2 are both supported in $K_T := K + [-T, T]$ for all $0 \leq t \leq T$. Since u_t^2 and u_x^2 are continuous on $\mathbb{R}$ (and thus on K_T) for all $0 \leq t \leq T$, by the extreme value theorem, there exist finite constants M and N so that for all $x \in K$ and $0 \leq t \leq T$, $|u_t|^2 \leq M\chi_{K_T}$ and $|u_x|^2 \leq N\chi_{K_T}$, where both dominating functions are integrable since K_T is compact. Thus using the Dominated Convergence Theorem, taking the derivative of $e(t)$ gives us,

$$\begin{aligned} \dot{e}(t) &= \frac{1}{2} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx = \frac{1}{2} \frac{\partial}{\partial t} \int_{K_T} u_t^2 + u_x^2 dx \\ &= \frac{1}{2} \int_{K_T} \partial_t(u_t^2 + u_x^2) dx \\ &= \int_{K_T} u_t u_{tt} + u_x \cdot \partial_x u_t dx \\ &= \int_{K_T} u_t(u_{tt} - u_{xx}) dx \quad (\text{by integration by parts}) \\ &= 0, \end{aligned}$$

where there is no boundary term since u_x, u_t are supported inside K_T . So we have shown that $\dot{e}(t) = 0$ for all $0 \leq t \leq T$. Since $T > 0$ was chosen arbitrarily, it follows that $\dot{e}(t) \equiv 0$. Therefore, the total energy is constant in t .

(ii) What we wish is that in the limit $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

First, we compute

$$u_t^2 - u_x^2 = h(x+t)h(x-t) - h(x+t)g'(x-t) + h(x-t)g'(x+t) - g'(x-t)g'(x+t).$$

Then

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_{-\infty}^{\infty} |h(y)h(y-2t)| dy + \int_{-\infty}^{\infty} |h(y)g'(y-2t)| dy \\ &\quad + \int_{-\infty}^{\infty} |h(y-2t)g'(y)| dy + \int_{-\infty}^{\infty} |g'(y-2t)g'(y)| dy. \end{aligned}$$

Let $K \subset \mathbb{R}$ be a compact set large enough that contains the support of both g and h . Then we may write,

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_K |h(y)h(y-2t)| dy + \int_K |h(y)g'(y-2t)| dy \\ &\quad + \int_K |h(y-2t)g'(y)| dy + \int_K |g'(y-2t)g'(y)| dy. \end{aligned}$$

We will only show that the first term on the right vanishes since the work for the remaining terms is identical. Since $h(u) \rightarrow 0$ as $|u| \rightarrow \infty$ (by compact support of h), we know that pointwise $|h(y)h(y-2t)| \rightarrow 0$ as $t \rightarrow \infty$. Next since h is continuous, there exists some finite $M > 0$ so that $|h(y)|, |h(y-2t)| \leq M$ for all $y \in K$. Thus, $|h(y)h(y-2t)| \leq M^2\chi_K \in L^1(\mathbb{R})$ since K is compact. Hence, by the Dominated Convergence Theorem, one has that the first term vanishes as $t \rightarrow \infty$. As we stated before, the remaining three terms also vanish as $t \rightarrow \infty$. Thus, we conclude that as $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

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Problem 2.1.